
Practical - 3

Aim:

To Implement of max-heap sort algorithm.

Theory:

The max-heap sort algorithm works by visualizing an array as a complete binary tree, restructuring it into a max-heap where the largest element sits at the root, and then repeatedly swapping that largest element to the end of the array

Algorithm

Algorithm for Max Heap Sort (10 Steps)

Start the algorithm.

Read the number of elements n and the input array.

Build a Max Heap from the given array.

Set the last element as the current heap boundary.

Swap the root element (maximum element) with the last element of the heap.

Reduce the heap size by one.

Apply Max Heapify to the root to restore the Max Heap property.

Repeat Steps 5–7 until the heap size becomes 1.

Display the sorted array in ascending order.

Stop the algorithm.

Algorithm for Min Heap Sort (10 Steps)

Start the algorithm.

Read the number of elements n and the input array.

Build a Min Heap from the given array.

Set the last element as the current heap boundary.

Swap the root element (minimum element) with the last element of the heap.

Reduce the heap size by one.

Apply Min Heapify to the root to restore the Min Heap property.

Repeat Steps 5–7 until the heap size becomes 1.

Reverse the array (if ascending order is required) or display it directly for descending order.

Stop the algorithm.

Program Code

```
#include <iostream>
#include <vector>
#include <cstdlib>
#include <ctime>
#include <chrono>
#include <algorithm>

using namespace std;
using namespace chrono;

//----- MAX HEAP -----//
void maxHeapify(vector<int> &arr, int n, int i)
{
    int largest = i;
    int left = 2 * i + 1;
```

```
int right = 2 * i + 2;

if (left < n && arr[left] > arr[largest])
    largest = left;

if (right < n && arr[right] > arr[largest])
    largest = right;

if (largest != i)
{
    swap(arr[i], arr[largest]);
    maxHeapify(arr, n, largest);
}
}

void maxHeapSort(vector<int> &arr)
{
    int n = arr.size();

    // Build Max Heap
    for (int i = n / 2 - 1; i >= 0; i--)
        maxHeapify(arr, n, i);

    // Heap Sort
    for (int i = n - 1; i > 0; i--)
    {
        swap(arr[0], arr[i]);
        maxHeapify(arr, i, 0);
    }
}

//----- MIN HEAP -----//
void minHeapify(vector<int> &arr, int n, int i)
{
    int smallest = i;
    int left = 2 * i + 1;
    int right = 2 * i + 2;
```

```
    if (left < n && arr[left] < arr[smallest])
        smallest = left;

    if (right < n && arr[right] < arr[smallest])
        smallest = right;

    if (smallest != i)
    {
        swap(arr[i], arr[smallest]);
        minHeapify(arr, n, smallest);
    }
}

void minHeapSort(vector<int> &arr)
{
    int n = arr.size();

    // Build Min Heap
    for (int i = n / 2 - 1; i >= 0; i--)
        minHeapify(arr, n, i);

    // Heap Sort
    for (int i = n - 1; i > 0; i--)
    {
        swap(arr[0], arr[i]);
        minHeapify(arr, i, 0);
    }

    // Reverse to get ascending order
    reverse(arr.begin(), arr.end());
}

//----- MAIN -----//
int main()
{
    int n;
```

```
cout << "Enter number of elements: ";
cin >> n;

vector<int> original(n);

srand(time(0));

for (int i = 0; i < n; i++)
    original[i] = rand() % 100000;

vector<int> maxHeapArray = original;
vector<int> minHeapArray = original;

//----- MAX HEAP SORT -----//
auto startMax = high_resolution_clock::now();

maxHeapSort(maxHeapArray);

auto endMax = high_resolution_clock::now();

//----- MIN HEAP SORT -----//
auto startMin = high_resolution_clock::now();

minHeapSort(minHeapArray);

auto endMin = high_resolution_clock::now();

//----- DISPLAY TIME -----//

auto nanoMax = duration_cast<nanoseconds>(endMax - startMax);
auto microMax = duration_cast<microseconds>(endMax - startMax);
auto milliMax = duration_cast<milliseconds>(endMax - startMax);
duration<double> secMax = endMax - startMax;

auto nanoMin = duration_cast<nanoseconds>(endMin - startMin);
auto microMin = duration_cast<microseconds>(endMin - startMin);
```

```
auto milliMin = duration_cast<milliseconds>(endMin - startMin);  
duration<double> secMin = endMin - startMin;
```

```
cout << "\n===== MAX HEAP SORT =====\n";  
cout << "Nanoseconds : " << nanoMax.count() << " ns\n";  
cout << "Microseconds : " << microMax.count() << " us\n";
```

```
cout << "\n===== MIN HEAP SORT =====\n";  
cout << "Nanoseconds : " << nanoMin.count() << " ns\n";  
cout << "Microseconds : " << microMin.count() << " us\n";
```

```
cout<<"Sangem Vijayendra Reddy\n";
```

```
cout<<"92460118132\n";  
cout<<"5-EN18\n";
```

```
    return 0;  
}
```

Result :

```
"C:\Users\Admin\Downloads\ X + v
Enter number of elements: 67

===== MAX HEAP SORT =====
Nanoseconds : 22600 ns
Microseconds : 22 us

===== MIN HEAP SORT =====
Nanoseconds : 23900 ns
Microseconds : 23 us
Sangem Vijayendra Reddy
92460118132
5-EN18
Process returned 0 (0x0) execution time : 4.189 s
Press any key to continue.
```

